UCAR PULSE

The Operating System for Tunisian Higher Education

HACK4UCAR 2025 · University of Carthage · ACM ENSTAB

> *"Today UCAR manages 30+ institutions with Excel files and email chains.*
> ***UCAR Pulse replaces that entire chain — from raw documents to strategic decisions."***

Role	Name	Responsibility
Backend — AI & API	Yassine (you)	Ingestion pipeline, KPI API, Anomaly Engine, Report Generation
Backend — Infrastructure	Bedi	EchoGarden → Supabase migration, AI Assistant, Multi-tenant setup
Frontend	Rayen	Dashboard, institution views, alerts, charts, rankings
Frontend + Pitch	Yassine F.	Chat UI, report center, ingest page, auth, PPT

1. The Problem We Are Solving

The University of Carthage oversees 30+ affiliated institutions. Every single one operates in isolation. Data lives in PDFs, scanned papers, Excel files, and WhatsApp messages. The result is institutional blindness at every level.

Who	Their Pain Today	How Long It Takes
UCAR Rector	Emails 30 institutions monthly, waits a week, copy-pastes numbers manually	3-4 weeks per cycle
Institution President	Prepares board reports by hand from scattered files	4+ hours per report
Finance Admin (Trésorier)	Tracks budget manually in Excel, no central validation	Full-time manual work
HR Admin (Chef Personnel)	Staff data scattered across paper registers and local files	No real-time visibility
Academic Admin (Scolarité)	Student KPIs compiled manually per semester	Weeks of delay

2. The Solution — UCAR Pulse

UCAR Pulse is a multi-tenant intelligent platform with two main dashboards: one for each Institution and one for UCAR central administration. AI handles extraction, monitoring, analysis, and report writing automatically.

The One-Line Pitch

Not a dashboard. Not a chatbot. The operating system for Tunisian higher education.

What Makes Us Different

- **We ingest ANY format — scanned PDF, photo of a printed table, Excel, Word. Others require a clean upload.**
- **Our AI does not hallucinate — EchoGarden's Weaver+Verifier returns citations and a pass/revise/abstain verdict.**
- **We generate real French reports — Groq writes the narrative, PDF downloads in 30 seconds.**
- **Two-dashboard architecture — Institution view + UCAR central view. Mirrors real Tunisian university governance.**
- **Achievements system — not just problems. Positive milestones are flagged and celebrated.**
- **Role-based access — mirrors actual Tunisian university roles (Président, Trésorier, Chef Personnel, Scolarité).**

3. The Two Dashboards

🏫 Dashboard 1 — Institution View

Each of UCAR's 30+ institutions has its own isolated space. Staff log in and see only their institution's data.

What Institutions Can Do

- Upload documents (PDFs, Excel, scanned reports) before the monthly deadline
- View their own KPIs across all 8 families — academic, finance, HR, research, etc.
- Download their monthly AI-generated report (available after deadline)
- Browse report history organized by month
- See their own alerts and achievements
- Ask the AI assistant questions about their own data only
- President approves/denies role access for staff members

Monthly Report Flow

```
Day 1-25 of month:  Institution uploads files (PDFs, Excel, scans)
Day 26:             Upload deadline — system locks submissions
Day 27-28:          AI processes all uploads → extracts KPIs → generates report
Day 29:             Monthly report available for download
                    ↓
If no files uploaded → Fallback report generated using last known data
                       with warning: 'Based on previous period data —
                       no new files received this month'
```

Fallback Behavior (No Upload)

If an institution misses the deadline, the system does not fail silently. It generates a report using the last available data with a visible warning banner, and UCAR receives an alert: 'INSAT has not submitted data for this period.'

🏛️ Dashboard 2 — UCAR Central View

The main feature. UCAR sees everything across all 30+ institutions in one consolidated view. This is where strategic decisions are made.

What UCAR Can Do

- See all institutions side by side with health status (green/orange/red per KPI family)
- View performance benchmarking across institutions — rankings visible only to UCAR
- Receive automatic critical alerts when any institution crosses a danger threshold
- Receive automatic achievement notifications when an institution hits a milestone
- Download the monthly consolidated report covering all institutions
- Request an urgent on-demand report for specific institutions mid-month
- Ask the AI assistant cross-institution questions ('Compare dropout trends across all institutions')
- See which institutions haven't submitted data and send reminders

Urgent Report Flow

```
UCAR clicks 'Urgent Report'
```

```
      ↓
Selects institution(s) + period
      ↓
Groq generates French narrative from latest available KPIs
      ↓
Anomaly section included automatically
      ↓
PDF ready in ~30 seconds
```

4. Role System — Based on Real Tunisian University Structure

Roles are based on the actual administrative structure of Tunisian public universities as defined by the Ministry of Higher Education.

Institution-Level Roles

Role (FR)	Role (EN)	What They See	What They Can Do
Président d'Institution	Institution President	Everything in their institution — all KPI families, all alerts, all reports	Approve/deny staff access, upload any document, download all reports, use AI assistant
Trésorier / Chef Service Budget	Finance Admin	Finance KPIs only: budget allocated, budget used, execution rate, cost per student, event costs, transport	Upload financial documents (budget reports, expense sheets), view finance alerts
Chef Service Personnel (RH)	HR Admin	HR KPIs only: staff headcount, absenteeism, training completion, contract types	Upload HR documents (staff registers, attendance sheets), view HR alerts
Chef Service Scolarité	Academic Admin	Academic KPIs only: success rate, dropout, attendance, exam results, enrollments	Upload academic documents (exam results, attendance registers), view academic alerts
Chef Service Recherche	Research Admin	Research KPIs only: publications, active projects, funding, patents, partnerships	Upload research documents, view research alerts

UCAR-Level Roles

Role	What They See	What They Can Do
Présidente de l'Université (UCAR)	Everything — all institutions, all KPIs, all rankings, all reports	Full access. Approves UCAR staff access. Requests urgent reports. Delegates access to VP in absence.
Vice-Président(e) Délégué(e)	Same as President when delegated	Same as President — activated only when President formally delegates
Secrétaire Général(e)	All institutions — administrative and financial KPIs	Download consolidated reports, view alerts, no ranking access

Access Approval Flow

```
New staff member registers on UCAR Pulse
      ↓
Selects their institution + role
      ↓
Account marked PENDING
      ↓
Institution President receives notification
      ↓
President approves or denies
      ↓
Staff member gets access (or rejection notification)
```

5. AI Features

🤖 AI Assistant (Powered by EchoGarden)

EchoGarden is a RAG (Retrieval-Augmented Generation) pipeline — not a chatbot. It ingests documents, stores them as vectors, and when a question is asked, it retrieves the most relevant context and generates a grounded answer. Every answer includes citations and a verification verdict.

Feature	Institution View	UCAR View
Data scope	Their institution only	All 30+ institutions
Question types	About their own KPIs and documents	Cross-institution comparison and analysis
Language	French, Arabic, English	French, Arabic, English
Answer quality	Grounded with citations — shows which document	Same quality, broader context
Verdict system	pass / revise / abstain — no hallucination	Same

Example Questions

Who Asks	Question	What the System Does
Institution President	Pourquoi notre taux d'abandon augmente ce semestre ?	Retrieves academic documents + KPI history, explains trend with citations
Finance Admin	Quel est notre taux d'exécution budgétaire par rapport au trimestre dernier ?	Pulls finance snapshots, calculates trend, cites source documents
UCAR	Quelle institution a le meilleur taux d'insertion professionnelle et pourquoi ?	Compares employment KPIs across all institutions, explains top performer
UCAR	Quelles institutions n'ont pas soumis leurs données ce mois-ci ?	Checks ingestion_jobs table, returns list with last submission dates

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🚨 Alert System

KPI	Threshold	Severity	Who Gets Notified
dropout_rate	> 20%	CRITICAL 🔴	Institution President + UCAR
attendance_rate	< 75%	WARNING 🟡	Institution President + UCAR
success_rate	< 60%	WARNING 🟡	Institution President + UCAR
budget_execution_rate	< 50%	WARNING 🟡	Trésorier + Institution President + UCAR
employability_rate	< 50%	CRITICAL 🔴	Institution President + UCAR
absenteeism_rate	> 8%	CRITICAL 🔴	Chef Personnel + Institution President + UCAR
training_completed_rate	< 50%	WARNING 🟡	Chef Personnel + UCAR
it_equipment_ok_rate	< 75%	WARNING 🟡	Institution President + UCAR
No data submitted	Monthly deadline passed	INFO 🔵	UCAR only

🏆 Achievement System

The platform is not only a problem detector. Positive milestones are flagged as achievements — visible to the institution and to UCAR. This creates a recognition culture, not just a surveillance system.

Achievement	Trigger	Example
Excellence Académique	success_rate > 85% for 2 consecutive periods	ESPRIT maintains 88% success rate
Insertion Professionnelle	employability_rate > 80%	ESPRIT reaches 84% employment rate
Bonne Gestion Budgétaire	budget_execution_rate between 85-100%	ENIT improves execution from 47% to 88%
Recherche Active	publications_count increases > 20% vs previous period	INSAT publishes 30% more papers
Hackathon / Compétition	Manually added by Institution President	INSAT wins HACK4UCAR 2025 🏆

6. Technical Stack

Layer	Technology	Used For
Frontend Framework	Next.js 14 (App Router)	Both dashboards — SSR, routing, performance
UI Styling	Tailwind CSS v3	All styling — no custom CSS
TypeScript	TypeScript v5	Type safety across frontend
Charts	Recharts	KPI trend charts, comparison bar charts
Backend Framework	FastAPI (Python 3.11)	All API endpoints — async, fast, auto Swagger docs
Database	Supabase (PostgreSQL 15)	All data — multi-tenant with institution_id
Authentication	Supabase Auth	Login, roles, JWT — row-level security
Vector Store	Qdrant	AI assistant semantic search
AI Pipeline	EchoGarden (Bedi)	Document ingestion, RAG, Weaver+Verifier
LLM	Groq — Llama 3.3 70b	KPI extraction, report narrative writing
OCR / Document Parsing	Apache Tika + Tesseract	Extract text from any file format
PDF Generation	WeasyPrint + HTML/CSS	Board reports and monthly reports
File Storage	Supabase Storage	Generated PDFs, uploaded documents
Containerization	Docker + Docker Compose	One command to run full stack

7. Database Schema

```
institutions      (id, name, slug, type, city, created_at)
users             (id, institution_id, email, role, status[pending/active/denied])
kpi_snapshots     (id, institution_id, period, data JSONB, source_file)
alerts            (id, institution_id, kpi_family, kpi_field, severity, message,
resolved)
achievements      (id, institution_id, type, title, description, period, created_at)
reports           (id, institution_id, type[monthly/urgent/consolidated], period,
pdf_url, status)
ingestion_jobs    (id, institution_id, filename, status, extracted_kpis, error_message)
memory_cards      (id, institution_id, content, qdrant_ref, source_type)  ← EchoGarden
conversation_turns(id, institution_id, user_message, answer, verdict, citations)
```

KPI Schema — All 8 Families (stored as JSONB in kpi_snapshots.data)

```
{
  "academic":       {"success_rate", "attendance_rate", "dropout_rate",
"repetition_rate", "exam_pass_rate"},
  "finance":        {"budget_allocated", "budget_used", "budget_execution_rate",
"cost_per_student"},
  "employment":     {"employability_rate", "insertion_delay_months",
"national_partnership_rate"},
  "hr":             {"teaching_staff_count", "admin_staff_count", "absenteeism_rate",
"training_completed_rate"},
  "research":       {"publications_count", "active_projects", "funding_secured",
"patents_filed"},
  "infrastructure": {"classroom_occupancy_rate", "it_equipment_ok_rate",
"ongoing_works"},
  "esg":            {"energy_consumption_kwh", "recycling_rate", "green_mobility_rate"},
  "partnerships":   {"active_agreements", "outgoing_mobility", "incoming_mobility",
"international_projects"}
}
```

8. Mock Data — The 4 Demo Institutions

Institution	Story	Main Alert	Achievement
INSAT	Dropout crisis — rising from 19% to 28% over 3 semesters. Attendance collapsing.	CRITICAL: Dropout 28%	None this period
ENIT	Great academics but only 47% budget execution. IT equipment deteriorating as a result.	WARNING: Budget execution 47%	None this period
ESPRIT	Star performer. Best employability, strongest research, clean finances. The benchmark.	None	Excellence Académique + Insertion Pro
ISG	HR crisis. Absenteeism at 13%, training at 29%. Academic results declining each semester.	CRITICAL: Absenteeism 13%	None this period

9. API Endpoints

Method	Endpoint	Description
GET	/healthz	Health check
GET	/institutions	List all institutions
GET	/institutions/:id/kpis	Latest KPI snapshot
GET	/institutions/:id/kpis/history	All periods — for trend charts
GET	/dashboard/summary	All institutions + alerts + health status
GET	/alerts	All alerts (filter: institution_id, severity, resolved)
POST	/alerts/:id/resolve	Mark alert as resolved
GET	/achievements	All achievements (filter: institution_id)
POST	/ingest/document	Upload file → Tika → Groq → KPI extraction → DB
GET	/ingest/jobs/:id	Check ingestion job status
POST	/reports/generate	Generate report (monthly or urgent)
GET	/reports	List reports (filter: institution_id, type)
GET	/reports/:id	Get report + PDF URL
POST	/chat	AI assistant query
GET	/rankings	Institution rankings by KPI family — UCAR only

10. Frontend Pages

Route	Page	Who Sees It	Key Components
/login	Login	Everyone	Email + password, role badge on login, pending state if not approved
/dashboard	Institution Dashboard	Institution staff	KPI family cards, active alerts, achievements, upload button, monthly report download
/dashboard/ucar	UCAR Dashboard	UCAR only	All institutions grid, health status, rankings, consolidated alerts, urgent report button
/ingest	File Upload	Institution staff (by role)	Drag-drop upload, institution + period selector, live extraction progress, KPI preview
/reports	Reports	Institution: own reports / UCAR: all	Monthly report history by month, download PDF, request urgent report
/chat	AI Assistant	All logged-in users	Chat interface, verdict badge, citations panel, context selector
/alerts	Alerts	Institution: own / UCAR: all	Alert list, severity filter, resolve button, institution filter
/admin	Access Management	Presidents only	Pending requests list, approve/deny buttons, active users list

11. The Demo Script — 6 Minutes

Min	What You Say	What You Show
0-1	Today the rector of UCAR manages 30 institutions with Excel and email. Every month: send emails, wait a week, receive 30 different formats, copy-paste manually. A crisis at INSAT can go unnoticed for weeks. We ended that loop.	Eye contact. No screen.
1-2	Let me show you what happens when INSAT's trésorier uploads their semester budget report.	Login as INSAT Trésorier. Navigate to /ingest. Upload scanned PDF. Watch extraction: 8 seconds, 12 KPIs appear. Alert fires automatically: Dropout 28% CRITICAL.
2-3	The UCAR rector opens her dashboard Monday morning.	Login as UCAR. Show 4 institutions. INSAT red, ENIT orange, ESPRIT green with achievement badge, ISG red. Click INSAT → dropout trend climbing 3 semesters.
3-4	She wants to understand the situation before the board meeting.	Navigate to /chat. Type in French: Pourquoi le taux d'abandon à l'INSAT augmente chaque semestre? Show grounded answer, citations, verdict: pass.
4-5	She needs a report in 2 hours. One click.	Navigate to /reports. Click Urgent Report, select INSAT. 30 seconds. PDF opens. AI-written French narrative, KPI table, anomaly section, 3 recommendations. Ask judge: how long does this take today?
5-6	UCAR Pulse is not a dashboard. It is the operating system for Tunisian higher education. 30 institutions. One platform. Zero Excel chains.	Return to UCAR dashboard. Slow pan across all 4 institutions. Stop on ESPRIT — achievement badge glowing. This is what success looks like.

12. How We Win Each Criterion

Criterion	How We Win
Impact	We kill the Excel chain. Every admin in the room recognizes this pain. Monthly reports, role-based access, fallback system — all show we understand the real workflow.
Innovation	EchoGarden Weaver+Verifier = no hallucination. Any-format ingestion. AI report in French in 30 seconds. Achievement system. Nobody else has all of this.
Usability	Role-based views — trésorier sees only finance. Approval flow mirrors real governance. French-language AI. Monthly deadline system with fallback. Built for non-technical university staff.
Scalability	Supabase multi-tenant: institution_id on every table. One deployment for all 30+ institutions. Row-level security enforced at DB level. Zero configuration per new institution.
Feasibility	FastAPI + Supabase + Groq = deployable today in Tunisia. No GPU. Groq free tier covers demo. Supabase free tier covers demo. Total infra cost at scale: under 100 USD/month.